DIFFERENTIAL EQUATIONS

(10 MARKS)

1% DIFFERENTIAL EQUATIONS , 99% INTEGRATION

$$y \underset{\text{integrate}}{\overset{\text{DIFF}}{\rightleftarrows}} \frac{dy}{dx}$$

DIFFERENTIAL EQ.

INTEGRATION

$$\frac{dy}{dx} = y^2 e^{3x}$$

$$dy = y^2 e^{3x}\, dx$$

$$\int \frac{1}{y^2}\, dy = \int e^{3x}\, dx$$

$$\int 1 \boxed{y^{-2}}\, dy = \frac{1}{3}\int 3\, e^{\boxed{3x}}\, dx$$

$$\frac{y^{-1}}{-1} = \frac{1}{3} e^{3x} + c$$

$$\boxed{-\frac{1}{y} = \frac{e^{3x}}{3} + c}$$

$x =$

$y =$

Find c.

7 Given that $y = 0$ when $x = 1$, solve the differential equation

$$xy\frac{dy}{dx} = y^2 + 4,$$

obtaining an expression for y^2 in terms of x. [6]

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DIFFERENTIAL EQ.

$$xy\frac{dy}{dx} = y^2+4$$

$$xy\,dy = (y^2+4)\,dx$$

$$\int \frac{y}{y^2+4}\,dy = \int \frac{1}{x}\,dx$$

INTEGRATION.

$$\frac{1}{2}\int \frac{2y}{\boxed{y^2+4}}\,dy = \int \frac{1}{\boxed{x}}\,dx$$

$\square = y^2+4$

$\square' = 2y$

$\square = x$

$\square' = 1$

$$\boxed{\frac{1}{2}\ln(y^2+4) = \ln x + c}$$

$y=0$

$x=1$

$$\frac{1}{2}\ln(0^2+4) = \ln 1 + c$$

$$\boxed{\frac{1}{2}\ln 4 = c}$$

SIMPLIFICATION

$$\left[\frac{1}{2}\ln(y^2+4) = \ln x + \frac{1}{2}\ln 4\right] \times 2$$

$$\ln(y^2+4) = 2\ln x + \ln 4$$

$$\ln(y^2+4) = \ln x^2 + \ln 4$$

$$\cancel{\ln}(y^2+4) = \cancel{\ln}(4x^2)$$

$$y^2 + 4 = 4x^2$$

$$\boxed{y^2 = 4x^2 - 4}$$

12 The variables x and y are related by the differential equation

$$\frac{dy}{dx} = \frac{6xe^{3x}}{y^2}.$$

It is given that $y = 2$ when $x = 0$. Solve the differential equation and hence find the value of y when $x = 0.5$, giving your answer correct to 2 decimal places. [8]

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$$\frac{dy}{dx} = \frac{6xe^{3x}}{y^2}$$

$$\int y^2\,dy = \int 6xe^{3x}\,dx$$

$$\int y^2\,dy = 6\int \underbrace{x}_{u}\,\underbrace{e^{3x}}_{v}\,dx$$

$$\frac{y^3}{3} = 6$$

Diff of u	Integ of v
$x \longrightarrow 1$	$\frac{1}{3}\int 3e^{3x}\,dx = \frac{1}{3}e^{3x} = \frac{e^{3x}}{3}$

$$u\int v\,dx - \int\left[\frac{du}{dx} \times \int v\,dx\right]dx$$

$$(x)\left(\frac{e^{3x}}{3}\right) - \int\left[1 \times \frac{e^{3x}}{3}\right]dx$$

$$\frac{xe^{3x}}{3} - \frac{1}{3}\int e^{3x}\,dx$$

$$\frac{y^3}{3} = 6\left[\frac{xe^{3x}}{3} - \frac{1}{3}\left(\frac{e^{3x}}{3}\right)\right] + c$$

$$\boxed{\frac{y^3}{3} = 2xe^{3x} - \frac{2e^{3x}}{3} + c}$$

$y = 2$, $x = 0$

$$\frac{2^3}{3} = 2(0)e^{3(0)} - \frac{2e^{3(0)}}{3} + c$$

$$\frac{8}{3} = -\frac{2}{3} + c$$

$$\boxed{c = \frac{10}{3}}$$

$$\boxed{\frac{y^3}{3} = 2xe^{3x} - \frac{2e^{3x}}{3} + \frac{10}{3}}$$

put $x = 0.5$

$$\frac{y^3}{3} = 2(0.5)e^{3(0.5)} - \frac{2}{3}e^{3(0.5)} + \frac{10}{3}$$

$y = \boxed{}$ 2dp.